

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12397960
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Couture; David G. et al.

Produce trays with reinforced tabs

Abstract

In accordance with at least one aspect of this disclosure, a blank for forming a container includes a base having a first pair of opposed base edges and a second pair of opposed base edges. The blank includes two first side wall structures, each first side wall structure foldably attached to one of the first base edges and two second side wall structures, each second side wall structure foldably attached to one of the second base edges.

Inventors: Couture; David G. (Hoschton, GA), McLeod; Michael (Romeoville, IL)

Applicant: WestRock Shared Services, LLC (Atlanta, GA)

Family ID: 1000008778542

Assignee: WestRock Shared Services, LLC (Atlanta, GA)

Appl. No.: 18/151388

Filed: January 06, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240228105 A1	Jul. 11, 2024

Publication Classification

Int. Cl.: B65D5/00 (20060101); B65D5/22 (20060101); B65D5/44 (20060101)

U.S. Cl.:

CPC B65D5/0045 (20130101); B65D5/22 (20130101); B65D5/443 (20130101);

Field of Classification Search

CPC: B65D (5/0045); B65D (5/22); B65D (5/443); B65D (5/0015); B65D (5/002); B65D (5/42); B65D (5/001)

USPC: 229/169; 229/187; 229/108; 229/176; 229/185.1; 229/919; 206/503; 206/509; 206/511; 206/557

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3102674	12/1962	Hamilton	229/187	B65D 5/48048
3114493	12/1962	Dunkin	229/117.17	B65D 5/22
5458283	12/1994	Southwell	N/A	N/A
8205787	12/2011	Panduro, Jr.	229/173	B65D 5/6629
8827142	12/2013	Smith	229/199	B31B 50/26
2002/0179481	12/2001	Noland	N/A	N/A
2014/0246484	12/2013	Armstrong et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
9210936	12/1991	DE	N/A
0575004	12/1992	EP	N/A

OTHER PUBLICATIONS

International Search Report for PCT/US23/85010 dated Apr. 26, 2021, 5 pages. cited by applicant

Primary Examiner: Demeree; Christopher R

Background/Summary

BACKGROUND

1. Field

(1) The present disclosure relates to containers for packaging product such as produce, and more particularly to tab structures that can be used for stacking multiple containers.

2. Description of Related Art

(2) Various designs exist for forming tray-like containers from corrugated paper material. Such containers can be used to hold product such as produce. It can be advantageous to stack multiple filled containers, one on top of the other, to transport the product and/or to save space in a retail setting. Stacking tabs from the containers can be used to interlock stacked containers together.

(3) The conventional techniques have been considered satisfactory for their intended purpose. However, there is an ever present need for improved systems and methods for improved tabs on stackable tray type containers. This disclosure provides a solution for this need.

SUMMARY

(4) In accordance with at least one aspect of this disclosure, a blank for forming a container includes a base having a first pair of opposed base edges and a second pair of opposed base edges. The blank includes two first side wall structures, each first side wall structure foldably attached to one of the first base edges and two second side wall structures, each second side wall structure

foldably attached to one of the second base edges. Each of the second side wall structures include a first wall panel foldably attached to one of the second base edges along a first fold line, a second wall panel foldably attached to the first wall panel configured to be overlapped with the first wall panel to form a side wall, a first minor end flap foldably attached to the first wall panel extending in a lateral direction from the first side wall panel, and a second minor end flap foldably attached to the second side wall panel extending in the lateral direction from the second side wall panel, configured to overlap the first minor end flap and one of the first side wall structures in forming a corner of a container formed from the blank.

(5) A stacking tab section is defined in the first and second minor end flaps and is configured to form a stacking tab extending upright from a first side wall of the container formed from the blank with the first and second minor end flaps overlapped with one of the side wall structures to form the first side wall. A first portion of the stacking tab section is formed in the first minor end flap and a second portion of the stacking tab section is formed in the second minor end flap. A cut separates the first and second portions of the stacking tab section, so the first and second portions of the stacking tab section can separate to facilitate erecting the container made from the blank. The first and second wall panels are configured to fold along a fold line to overlap the first and second wall panels with one another to form the first wall panel to form a side wall.

(6) In embodiments, the first and second wall panels can be configured to form a first end wall adjacent to the first side wall with the corner of the container connecting between the first end wall and the first side wall.

(7) In certain embodiments, the stacking tab section can be a first stacking tab section. A first portion of a second stacking tab section can be defined on the first wall panel and a second portion of the second stacking tab section can be defined on the second wall panel where the fold line can connect the first and second portions of the second stacking tab section. The second stacking tab section can be configured to form a second stacking tab extending upward from the first end wall of the container formed from the blank.

(8) In certain embodiments, a void can separate the first and second wall panels so that in the container formed from the blank, the fold line is configured to form a top edge of the second stacking tab, relative to the base. One or more edges of the void can form top edges of the first end wall and of the first side wall below the top edge of the second stacking tab, relative to the base.

(9) In certain embodiments, the first minor end flap can include a first corner panel and the second minor end flap can include a second corner panel. The first and second corner panels can be configured to overlap and form a beveled wall connecting the first end wall to the first side wall in the container formed from the blank.

(10) In certain embodiments, the first and second minor end flaps can be long enough meet opposing first and second minor end flaps at a midpoint of the base in the container formed from the blank. The first stacking tab section can be configured to form a single stacking tab in cooperation with an opposing stacking tab section of the opposing first and second minor end flaps.

(11) In certain embodiments, the first and second minor end flaps are not long enough meet opposing first and second minor end flaps at a midpoint of the base in the container formed from the blank. The first stacking tab section can be configured to form a separate stacking tab from an opposing stacking tab section of the opposing first and second minor end flaps.

(12) In certain embodiments, a first portion of a third stacking tab section can be defined on the first wall panel and a second portion of the third stacking tab section can be defined on the second wall panel. The fold line can connect the first and second portions of the third stacking tab section and the third stacking tab section can be configured to form a third stacking tab extending upward from the first end wall of the container formed from the blank.

(13) In certain embodiments, the base can define an anti-nesting corner configured to extend outside the first end wall and the first side wall in the container formed from the blank.

(14) In certain embodiments, the first minor end flap can be foldably connected directly to the first

wall panel, and the second minor end flap can be foldably connected directly to the second wall panel so that in the container formed from the blank, the first end wall is foldably connected directly to the first side wall without an intervening bevel panel.

(15) In certain embodiments, the first and second minor end flaps can be long enough meet opposing first and second minor end flaps at a midpoint of the base in the container formed from the blank. The first stacking tab section can be configured to form a single stacking tab in cooperation with an opposing stacking tab section of the opposing first and second minor end flaps.

(16) In certain embodiments, the first and second minor end flaps are not long enough meet opposing first and second minor end flaps at a midpoint of the base in the container formed from the blank. The first stacking tab section can be configured to form a separate stacking tab from an opposing stacking tab section of the opposing first and second minor end flaps.

(17) In certain embodiments, a first portion of a third stacking tab section can be defined on the first wall panel and a second portion of the third stacking tab section can be defined on the second wall panel. The fold line can connect the first and second portions of the third stacking tab section and the third stacking tab section can be configured to form a third stacking tab extending upward from the first end wall of the container formed from the blank.

(18) In accordance with at least aspect of this disclosure, a container can be made from a blank as described herein above, where each of the first side wall structures can be positioned inward from the first and second wall panels.

(19) In accordance with at least aspect of this disclosure, a container can be made from a blank as described herein above, where each of the first sidewall structures can be positioned between first and second minor end flaps of the second sidewall structures.

(20) These and other features of the systems and methods of the subject disclosure will become more readily apparent to those skilled in the art from the following detailed description of the preferred embodiments taken in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that those skilled in the art to which the subject disclosure appertains will readily understand how to make and use the devices and methods of the subject disclosure without undue experimentation, preferred embodiments thereof will be described in detail herein below with reference to certain figures, wherein:

(2) FIG. 1A is a plan view of an embodiment of a blank constructed in accordance with the present disclosure, showing stacking tab portions for forming stacking tabs in a container;

(3) FIG. 1B is a perspective view of a container constructed from the blank of FIG. 1A, showing the stacking tabs;

(4) FIG. 2A is a plan view of an embodiment of a blank constructed in accordance with the present disclosure, showing anti-nesting features;

(5) FIG. 2B is a perspective view of a container constructed from the blank of FIG. 2A, showing the anti-nesting features;

(6) FIG. 3A is a plan view of an embodiment of a blank constructed in accordance with the present disclosure, showing stacking tab features on corner folds;

(7) FIG. 3B is a perspective view of a container constructed from the blank of FIG. 3A, showing the stacking tabs on the corner folds;

(8) FIG. 4A is a plan view of an embodiment of a blank constructed in accordance with the present disclosure, showing stacking tab portions for end tabs;

(9) FIG. 4B is a perspective view of a container constructed from the blank of FIG. 4A, showing the end tabs;

(10) FIG. 5A is a plan view of an embodiment of a blank constructed in accordance with the present disclosure, showing stacking tab portions for additional end tabs;

(11) FIG. 5B is a perspective view of a container constructed from the blank of FIG. 5A, showing the additional end tabs;

(12) FIG. 6A is a plan view of an embodiment of a blank constructed in accordance with the present disclosure, showing a rectangular base for forming a rectangular container;

(13) FIG. 6B is a perspective view of a container constructed from the blank of FIG. 6A, showing the wall panels in a rectangular configuration around the base;

(14) FIG. 7A is a plan view of an embodiment of a blank constructed in accordance with the present disclosure, showing the first sidewall structures configured to be nested between the minor end flaps;

(15) FIG. 7B is a perspective view of a container constructed from the blank of FIG. 7A, showing the first sidewall structures nested between the minor end flaps;

(16) FIG. 8 is a perspective view of a container constructed from a blank similar to that in FIG. 7A, but with the first sidewall structures fully inward from the minor end flaps;

(17) FIG. 9 is a perspective view of a container constructed from a blank similar to that in FIG. 7A, but with the first sidewall structures fully outward from the minor end flaps;

(18) FIG. 10A is a plan view of an embodiment of a blank constructed in accordance with the present disclosure, showing fold lines along which the first and second wall panels are directly connected;

(19) FIG. 10B is a perspective view of a container constructed from the blank of FIG. 10A, showing the stacking tabs;

(20) FIG. 11A is a plan view of an embodiment of a blank constructed in accordance with the present disclosure, showing stacking tab portions along corner fold lines;

(21) FIG. 11B is a perspective view of a container constructed from the blank of FIG. 11A, showing the stacking tabs formed along the corner fold lines;

(22) FIG. 12A is a plan view of an embodiment of a blank constructed in accordance with the present disclosure, showing cut top edges of the stacking tab portions;

(23) FIG. 12B is a perspective view of a container constructed from the blank of FIG. 12A, showing the stacking tabs with cut top edges;

(24) FIG. 13A is a plan view of an embodiment of a blank constructed in accordance with the present disclosure, showing a similar blank to that of FIG. 12A but with additional stacking tab portions for additional end tabs;

(25) FIG. 13B is a perspective view of a container constructed from the blank of FIG. 13A, showing the additional end tabs;

(26) FIG. 14A is a plan view of an embodiment of a blank constructed in accordance with the present disclosure, showing a rectangular version similar to the octagonal blank of FIG. 13A; and

(27) FIG. 14B is a perspective view of a container constructed from the blank of FIG. 14A, showing the rectangular shape of the container.

(28) FIG. 15A is a plan view of an embodiment of a blank constructed in accordance with the present disclosure, showing stacking tab portions separated by a cut;

(29) FIG. 15B is a perspective view of a container constructed from the blank of FIG. 15A, showing the end tabs;

(30) FIG. 16A is a plan view of an embodiment of a blank constructed in accordance with the present disclosure, similar to the blank of FIG. 15A but where the stacking tab portions are long enough to meet at a middle of a base of the container;

(31) FIG. 16B is a perspective view of a container constructed from the blank of FIG. 16A;

(32) FIG. 17A is a plan view of an embodiment of a blank constructed in accordance with the present disclosure, similar to the blank of FIG. 16A but having U shaped features in one or more of the side walls and/or the stacking tab portions;

- (33) FIG. 17B is a perspective view of a container constructed from the blank of FIG. 17A;
- (34) FIG. 18A is a plan view of an embodiment of a blank constructed in accordance with the present disclosure, similar to the blank of FIG. 17A but being configured to form a rectangular container;
- (35) FIG. 18B is a perspective view of a container constructed from the blank of FIG. 18A;
- (36) FIG. 19A is a plan view of an embodiment of a blank constructed in accordance with the present disclosure, similar to the blank of FIG. 5A, showing stacking tab portions as end tabs, the stacking tab portions separated by a cut and having an additional panel on each sidewall structure for forming the container by hand assembly;
- (37) FIG. 19B is a perspective view of a container constructed from a blank of FIG. 19A;
- (38) FIG. 20A is a plan view of an embodiment of a blank constructed in accordance with the present disclosure, similar to the blank of FIG. 19A but being configured to form a rectangular container;
- (39) FIG. 20B is a perspective view of a container constructed from a blank of FIG. 20A;
- (40) FIG. 21A is a plan view of an embodiment of a blank constructed in accordance with the present disclosure, similar to the blank of FIG. 19A but without the additional panel on the sidewall structure, and further being configured to form a rectangular container; and
- (41) FIG. 21B is a perspective view of a container constructed from the blank of FIG. 21A.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(42) Reference will now be made to the drawings wherein like reference numerals identify similar structural features or aspects of the subject disclosure. For purposes of explanation and illustration, and not limitation, a partial view of an embodiment of a blank in accordance with the disclosure is shown in FIGS. 1A and 1s designated generally by reference character **100**. Other embodiments of systems in accordance with the disclosure, or aspects thereof, are provided in FIGS. 1B-22, as will be described. The systems and methods described herein can be used to provide blanks to make stackable containers with stacking tabs for containing product such as produce. Blanks as disclosed herein can be configured to be erected into containers by machine or manually or both.

(43) The blank **100** for forming a stackable container **101** (shown in FIG. 1B) includes a base **102** having a first pair of opposed base edges **104**, **106** and a second pair of opposed base edges **108**, **110**. Two first sidewall structures **112**, **114** are each foldably attached to one of the first base edges **104**, **106**, respectively. Two second sidewall structures **116**, **118** are each foldably attached to one of the second base edges **108**, **110**, respectively.

(44) Each of the second sidewall structures **116**, **118** includes a first wall panel **120** foldably attached to one of the second base edges **108**, **110** along a first fold line, which lies along its respective base edge **108**, **110**. A respective second wall panel **122** is foldably attached to each first wall panel **120** configured to be overlapped with the respective first wall panel **120** to form a respective side wall **124** (labeled in FIG. 1B).

(45) A respective first minor end flap **126** is foldably attached to each end of each of the first wall panels **120** (for a total of four minor end flaps **126**) extending in a lateral direction from its respective first side wall panel **120**. A second minor end flap **128** is foldably attached to each end of each of the second side wall panels **122**, extending in the lateral direction from its respective second side wall panel **122**. There are at total of four minor end flaps **126** and four minor end flaps **128**. The minor end flaps **128** are configured to overlap the first minor end flaps **126**, respectively, and a proximate one of the first sidewall structures **112**, **114** in forming a corner **130** (labeled in FIG. 1B) of a container **101** formed from the blank **100**.

(46) A stacking tab section **132** is defined in the first and second minor end flaps **126**, **126** that is configured to form a stacking tab **134**, shown in FIG. 1B extending upright from a first side wall **136** of the container **101** formed from the blank **100** with the first and second minor end flaps **126**, **128** overlapped with one of the side wall structures **112**, **114** to form the first side wall **136**. To form the container **101** of FIG. 1B from the blank of FIG. 1A, the second sidewall structures **116**,

118 are both folded upright along their respective base edges **108, 110**, then the second side wall panels **122** are folded downward into overlapping position with the first side wall panels **120** to form the sidewalls **124** on either end of the container, which puts the second side wall panels **122** on the inside of the container **101**. The first sidewall structures **112, 114** are folded upright along their respective base edges **104, 106**. Then the minor end flaps **126, 128**, which are already overlapped with one another at each respective corner **130**, are folded and adhered against their adjacent sidewall structure **112, 114**.

(47) This process of erecting the container **101** can be performed manually, or by machine, and those skilled in the art will readily appreciate that any suitable variation in the steps above can be used without departing from the scope of this disclosure. The fold lines in FIG. **1A** are indicated with broken lines, and the thicker broken lines indicate crush lines, wider than normal fold lines, that facilitate forming the blank **100** into the container **100**, e.g. if the blank **100** is of a corrugated paper material or the like. Similar fold and crush line markings are used throughout the Figures.

(48) With continued reference to FIGS. **1A** and **1B**, the first and second wall panels are configured to form a first **124** adjacent to the first side walls **136** with each corner **130** of the container **101** connecting between adjacent end walls **124** and side wall **136**. A first portion **138** of each stacking tab section **132** is formed in the respective first minor end flap **126**. A second portion **140** of the stacking tab section **132** is formed in the second minor end flap **128**. A fold line **142** connects the first and second portions **138, 140** of each stacking tab section **132**. The first and second wall panels **120, 122** are configured to fold along the fold line, **142** to overlap the first and second wall panels **120, 122** with one another to form each first wall panel to form side walls, specifically the end walls **124**.

(49) A void **144**, labeled in FIG. **1A**, separates the first and second wall panels **120, 122** in each second sidewall structure **116, 118** so that in the container **101** of FIG. **1B**, the fold line **142** is on a top edge of each of the stacking tabs **134** formed from the stacking tab sections **132**, relative to the base **102**, i.e. as oriented in FIG. **1B**. Edges **146** of the void **144** form top edges of the end walls **124** and of the side walls **136** below the top edge of the stacking tabs **134**, relative to the base. The second side wall structures **116, 118** of FIG. **1A** are configured so that there is no stacking tab **134** extending upward from the end walls **124** in the container **100** of FIG. **1B**.

(50) With continued reference to FIGS. **1A** and **1B**, each first minor end flap **126** includes a first corner panel **148**. Each second minor end flap **128** includes a second corner panel **150**, wherein the pairs of first and second corner panels **148, 150** are each configured to overlap and form a beveled wall **152** connecting the first end walls **124** to the first side walls **136** in the container **101** of FIG. **1B**.

(51) FIGS. **2A** and **2B** show a similar blank **200** and corresponding container **201** to those shown in FIGS. **1A** and **1B**. However, the base **202** of the blank **200** and container **201** includes anti-nesting features **254** in its corners. These anti-nesting features **254** are optional and can make it easier to keep the containers **201** from nesting down into one another when stacking them together, e.g. when full of product.

(52) With reference now to FIGS. **3A** and **3B**, another blank **300** and corresponding container **301** are shown similar to those in FIGS. **1A** and **1B**. However, a portion each of the stacking tab sections **332** is formed in the first and second corner panels **348, 350**. The result is that in the container **301** formed from the blank **300**, the stacking tab sections **332** forms stacking tabs **334** extending along both sides of a fold line **356** connecting the first sidewalls **336** and the first end wall **324**.

(53) With reference now to FIGS. **4A** and **4B**, another blank **400** container **401** are shown that are similar to those in FIGS. **1A** and **1B**. However, additional stacking tab sections **458** defined in the first and second wall panels **420, 422**, configured to form a stacking tab, i.e. end tabs **460**, extending upright from the end walls **424** of the container **401** formed from the blank **400**. The corners **460** of the container **400** connect between the end walls **424** and the side wall **436**. The end

tabs **460** are in addition to the stacking tabs **434** that are similar to those described above for FIGS. **1A** and **1B**, so there are a total of six stacking tabs **434**, **460** on the container **401**.

(54) The first and second stacking tab portions (not numbered in FIG. **4A** to avoid obscuring the Figure, but see reference character **132** in FIG. **1A**) each fold along the fold line **442**. The fold line **442** is interrupted between the first and second stacking tab portions by the voids **444**, which are similar to the voids **144** in FIG. **1A**, but the stacking tab sections **458** interrupt the voids **444** making for a total of four voids **444**.

(55) With reference now to FIGS. **5A** and **5B**, another blank **500** is shown for forming another octagonal container **501** much as the octagonal containers **101**, **201**, **301**, **401** described above. However, the minor end flaps (see reference characters **126**, **128** in FIG. **1A**) are made long enough to merge the two stacking tabs **534** on each side wall **536** into a single stacking tab **534** so there are only two stacking tabs **534** on each side wall, as opposed to the four stacking tabs **434** in FIG. **4B**. In addition, the blank **500** includes two additional stacking tab portions **558** for a total of four end tabs **560** in the container **501**, as opposed to the two end tabs **460** of FIG. **4B**. The container **501** has a total of six stacking tabs **534**, **560**. Also note that the folding order for the container **501** is changed from the foregoing containers **101**, **201**, **301**, **401** so that the first sidewall structures **512**, **514** are folded up last and adhered to the outer surface of the minor end flaps (not numbered in FIGS. **5A** and **5B**, but see reference characters **126** in FIGS. **1A** and **1B**) rather than on the inside of the container **500**. It is, however, possible to form the container **500** with the first sidewall structures **512**, **514** on the inside of the container **500** as shown, e.g. in FIG. **1B**.

(56) With reference now to FIGS. **6A** and **6B**, there is shown a blank **600** and corresponding container **601** similar to that shown in FIGS. **4A** and **4B**. However, unlike the foregoing octagonal containers **100**, **200**, **300**, **400**, **500**, the container **600** is rectangular. The first minor end flaps **626** are each foldably connected directly to the respective first wall panels **620**. The second minor end flaps **628** are each foldably connected directly to the respective second wall panels **622** so that in the container **601** formed from the blank **600**, the end walls **624** are foldably connected directly to the side walls **363** without an intervening bevel panel such as the panels **148**, **150** in FIGS. **1A** and **1B**. The minor end flaps **626** and **628** are long enough to meet each other in the middle of the sidewalls **636**, covering the first sidewall structures **612**, **614** which are on the inside of the container **601**. However, the minor end flaps **626** and **628** could be made shorter to expose the outward facing surface of the first sidewall structures **612**, **614**, as in FIGS. **1B** and **4B**. The stacking tabs **634** and **660** are configured similar to those in FIG. **4B**.

(57) With reference now to FIGS. **7A** and **7B**, there is shown a blank **700** and corresponding rectangular container **701** similar to that shown in FIGS. **6A** and **6B**, but with a configuration of stacking tabs **736**, **760** similar to that in FIGS. **5A** and **5B**. Each of the first sidewall structures **712**, **714** is positioned inward of first minor end flaps **726** and outward of second minor end flaps **728**. However, it is also possible to form the container **701** with the first sidewall structures **712**, **714** on the inside of the container **701**, inward of the first and second minor end flaps **726**, **728** as shown in FIG. **8**, fully outside of the first and second minor end flaps **726**, **728** as shown in FIG. **9**.

(58) In the foregoing containers and blanks, the first and second wall panels fold on fold lines, e.g. fold lines **142** in FIGS. **1A** and **1B**, that form the top edges of the stacking tabs, e.g. stacking tabs **134** of FIG. **1B**. Now, with reference to FIGS. **10A** through **14B**, there are blanks **1000**, **1100**, **1200**, **1300**, **1400** and their containers **1001**, **1101**, **1201**, **1301**, **1401**, with a different folding configuration.

(59) With reference now to FIGS. **10A** and **10B**, there is a blank **1000** and corresponding octagonal container **1001** similar in most respect to those in FIGS. **1A** and **1B**, but for how the first and second wall structures **1016**, **1018**. The first and second wall panels **1020**, **1022** are configured to fold along a first fold line **1042**, which can include any suitable combination of scores, perforations, intermittent cuts, or the like, to overlap the first and second wall panels **1020**, **1022** with one another to form the first side walls, namely the end walls **1024** of the container **1001** in

FIG. 10B. The fold line **1042** directly connects between the first and second wall panels **1020**, **1022** of each wall structure **1016**, **1018** to form a top edge of the end walls **1024** relative to the base **1002**. A second fold line **1043** connects first and second portions **1038**, **1040** of each of the stacking tab sections **1032**. The second fold line **1043** is spaced apart farther from the base **1002** than is the first fold line **1042** in each wall structure **1016**, **1018**, so in a container **1001** formed from the blank **1000**, the second fold line **1043** forms a top edge of the stacking tabs **1034** formed from the first and second portions **1038**, **1040** of the stacking tab sections **1032**. In the container **1000** of FIG. 10B, the second fold line **1043** is above the first fold line **1042**, relative to the base **1002**, i.e. as oriented in FIG. 10B. It can be seen that in container **1001** there are no end tabs such as the end tabs **460** in FIG. 4B.

(60) With reference now to FIGS. 11A and 11B, there is a blank **1100** and corresponding container **1101** similar to those shown in FIGS. 10A and 10B. However, the stacking tabs **1134** in the container **1101** of FIG. 11B are formed on a corner fold, much as shown and described for FIGS. 3A and 3B. The first and second wall structures **1112**, **1114** include an additional corner panel **1113** on each end, which can be optionally omitted as it is in FIGS. 10A and 10B.

(61) As shown in FIGS. 12A and 12B, a blank **1200** and corresponding container **1201** are shown similar to those of FIGS. 11A and 11B. However, the stacking tab sections **1232** are each defined by a cut **1233** in the first and second minor end flaps **1226**, **1228**, so that in the container **1201** formed from the blank **1200**, a top edge of each stacking tab **1234** formed of from the stacking tab section **1232** is defined by the cut **1233**.

(62) FIGS. 13A and 13B show a blank **1300** and corresponding container similar to those of FIGS. 11A and 11B, but they include stacking tab portions **1358** for forming end tabs **1360** similar those described above for FIGS. 4A and 4B. FIGS. 14A and 14B show a blank **1400** and corresponding container **1401** with similar stacking tabs **1434** to shows shown in FIG. 10B, similar end tabs **1460** to those shown in FIG. 13B, but in a rectangular configuration, similar to the container **600** FIG. 6B.

(63) In accordance with at least one aspect of this disclosure, with reference now to FIGS. 15A and 15B, a blank **1500** for forming a corresponding container **1501** can be similar to the blank **400** and container **401** as shown and described in FIGS. 4A and 4B. The blank **1500** can include stacking tab section **1532** defined in the first and second minor end flaps **1526**, **1528** to form a stacking tab **1534**. The first portion **1538** of each stacking tab second **1532** is formed in the respective first minor end flap **1526** and the second portion **1540** of each stacking tab section **1532** is formed in the respective second minor end flap **1528**. Unlike blank **400**, in blank **1500**, a cut **1542** separates the first and second portions of the stacking tab section **1532**, so the first and second portions **1538**, **1540** of the stacking tab section **1532** can separate to facilitate erecting the container **1501** made from the blank **1500**. Accordingly, there is no fold line connecting the respective stacking tab portions **1538**, **1540** in the erected container **1501**.

(64) Still with reference to FIGS. 15A and 15B, the first and second minor end flaps **1526**, **1528** are not long enough meet opposing first and second minor end flaps at the middle of the base **1502** in the container **1501**, thus the first stacking tab section **1532** can be configured to form a separate stacking tab **1534** from an opposing stacking tab section **1542** of the opposing first and second minor end flaps **1526**, **1528**. The blank **1500** can include second stacking tab sections **1558** configured to form a second stacking tab (e.g., end tabs **1560**). In blank **1500**, the second stacking tab sections **1558** can be connected via fold lines.

(65) In certain embodiments, such as shown in FIGS. 16A and 16B, a blank **1600** for forming corresponding container **1601** can be similar to that of blank **1500** and container **1501**. In blank **1601**, the first and second minor end flaps **1626**, **1628** can be long enough meet opposing first and second minor end flaps at a middle of the base **1602** in the container **1601** formed from the blank **1600**. Here, the first stacking tab section **1632** can be configured to form a single stacking tab **1634** in cooperation with an opposing stacking tab section **1632** of the opposing first and second minor

end flaps **1626**, **1628**.

(66) In certain embodiments, with reference now to FIGS. **17A** and **17B** a blank **1701** for forming container **1701** can be similar to that of blank **1601** except that the respective wall structures **1712**, **1714** and the stacking tab sections **1732** can each include one or more U shaped features **1764** defined therein. The U shaped **1764** features of the walls structures **1712**, **1714** can correspond to the U shaped features **1764** of the respective stacking tab sections **1732** in the formed container **1701**. In certain embodiments, the innermost flaps **1766** of the respective wall structures **1712**, **1714** can be omitted.

(67) In certain embodiments, with reference now to FIGS. **18A** and **18B**, a blank **1800** for forming corresponding container **1801** can be similar to the blank **1700** and container **1701** as shown in FIGS. **17A** and **17B**. However, in blank **1800** the corresponding container **1801** formed can be rectangular and the U shaped features **1864** can also be rectangular. The beveled edges can be omitted to form the rectangular container **1801** similar to that of container **601** shown and described with respect to FIGS. **6A** and **6B**. The stacking tab sections **1834** can have either a fold line (e.g., as shown in the container **1801**) or a cut **1844** (e.g., as shown in the blank **1800**).

(68) With reference now to FIGS. **19A** and **19B**, the blank **1900** and container **1901** can be similar to blank **500** and container **501** shown in FIGS. **5A** and **5B**. In container **1901**, the stacking tabs **2034** formed in the respective side walls **2012**, **2014** have a cut **2044** separating the respective stacking tab portions **2026**, **2028**, instead of a fold line. In addition, to aid in erecting the container **1901** by hand, the sidewall structures **1912**, **1914** can have an additional panel **1968**, **1970** configured to fold over the sidewall structure, locking the sidewall structures **1912**, **1914** in place with a respective locking tab **1972**, **1974** inserted into slots **1976**, **1978**. When erecting the container **1901** from blank **1900**, the sidewall structures **1912**, **1914** can be folded upward, then the end walls and stacking tab sections **1932** are folded up and stacking tab sections **1932** are folded around sidewall structures **1912**, **1914**. The additional panels **1968**, **1970** are folded overtop of the stacking tab sections **1932** and the stacking tabs **1934** extend through a slot **1980**, **1982** while locking tabs **1972**, **1974** are inserted into slots **1976**, **1978**, holding the container **1901** in the erected state without any adhesive. This allows for in the field construction of the container **1901** by hand and without machine assistance. FIGS. **21A** and **20B** show a blank **2000** for forming container **2001** which can be similar to the blank **1900** and container **1901**, but blank **2000** can be designed to form a rectangular container **2001**, whereas container **1901** is octagonal.

(69) With reference now to FIGS. **21A-21B**, a blank **2100** for forming corresponding container **2101** can be similar to that of blank **2000** and container **2001** except that the blank **2100** can omit the additional sidewall panels **2068**, **2070**. The respective stacking tab portions **2126**, **2128** can be shorter than in blank **2000** such that when in the formed container **2100**, a larger gap between the opposed stacking tab portions **2126** is formed than in container **2000**.

(70) One having ordinary skill in the art, in view of this disclosure, would readily appreciate that one or more embodiments of the blanks and containers shown and described herein can have a folding sequence such that the first side wall structures can be positioned inward from the first and second wall panels in the erected container (e.g., as shown in at least FIGS. **15A-16B**), or such that sidewall structures can be positioned between first and second minor end flaps of the second sidewall structures in the erected container. In embodiments, the formed containers can have minor end flaps positioned as the inner most panel of the formed container (e.g., as shown in at least FIGS. **17A-18B**), or in embodiments, the minor end flaps can be positioned between respective stacking tab portions (e.g., as shown in at least FIGS. **21A-21B**).

(71) The methods and systems of the present disclosure, as described above and shown in the drawings, provide for stackable containers with stacking tabs for containing product such as produce. While the apparatus and methods of the subject disclosure have been shown and described with reference to preferred embodiments, those skilled in the art will readily appreciate that

changes and/or modifications may be made thereto without departing from the scope of the subject disclosure.

Claims

1. A blank for forming a container comprising: a base having a first pair of opposed base edges and a second pair of opposed base edges; two first side wall structures, each first side wall structure foldably attached to one of the first base edges; and two second side wall structures, each second side wall structure foldably attached to one of the second base edges, wherein each of the second side wall structures comprises: a first wall panel foldably attached to one of the second base edges along a first fold line, a second wall panel foldably attached to the first wall panel configured to be overlapped with the first wall panel to form a side wall, and a first minor end flap foldably attached to the first wall panel extending in a lateral direction from the first side wall panel, and a second minor end flap foldably attached to the second side wall panel extending in the lateral direction from the second side wall panel, configured to overlap the first minor end flap and one of the first side wall structures in forming a corner of a container formed from the blank, wherein a stacking tab section is defined in the first and second minor end flaps that is configured to form a stacking tab extending upright from a first side wall of the container formed from the blank with the first and second minor end flaps overlapped with one of the side wall structures to form the first side wall, wherein a first portion of the stacking tab section is formed in the first minor end flap, wherein a second portion of the stacking tab section is formed in the second minor end flap, and wherein a fold line connects the first and second portions of the stacking tab section, wherein the first and second wall panels are configured to fold along the fold line to overlap the first and second wall panels with one another to form the first wall panel to form a side wall.
2. The blank as recited in claim 1, wherein the first and second wall panels are configured to form a first end wall adjacent to the first side wall with the corner of the container connecting between the first end wall and the first side wall.
3. The blank as recited in claim 2, wherein the second side wall structures are configured so that there is no stacking tab extending upward from the first end wall in the container formed from the blank.
4. The blank as recited in claim 3, wherein the first minor end flap includes a first corner panel, wherein the second minor end flap includes a second corner panel, wherein the first and second corner panels are configured to overlap and form a beveled wall connecting the first end wall to the first side wall in the container formed from the blank.
5. The blank as recited in claim 4, wherein a portion of the stacking tab section is formed in the first and second wall panels, so in the container formed from the blank the stacking tab section forms a stacking tab extending along both sides of a fold line connecting the first sidewall and the first end wall.
6. The blank as recited in claim 1, wherein a void separates the first and second wall panels so that in the container formed from the blank, the fold line is on a top edge of the stacking tab formed from the stacking tab section, relative to the base.
7. The blank as recited in claim 6, wherein the second side wall structures are configured so that there is no stacking tab extending upward from the first end wall in the container formed from the blank.
8. The blank as recited in claim 7, wherein the first minor end flap includes a first corner panel, wherein the second minor end flap includes a second corner panel, wherein the first and second corner panels are configured to overlap and form a beveled wall connecting the first end wall to the first side wall in the container formed from the blank.
9. The blank as recited in claim 8, wherein a portion of the stacking tab section is formed in the first and second wall panels, so in the container formed from the blank the stacking tab section

forms a stacking tab extending along both sides of a fold line connecting the first sidewall and the first end wall.

10. The blank as recited in claim 7, wherein the first minor end flap is foldably connected directly to the first wall panel, and wherein the second minor end flap is foldably connected directly to the second wall panel so that in the container formed from the blank, the first end wall is foldably connected directly to the first side wall without an intervening bevel panel.

11. The blank as recited in claim 6 wherein edges of the void form top edges of the first end wall and of the first side wall below the top edge of the stacking tab, relative to the base.

12. The blank as recited in claim 1, wherein the stacking tab section is a first stacking tab section and further comprising: a second stacking tab section defined in the first and second wall panels, configured to form a stacking tab extending upright from a first end wall of the container formed from the blank with the corner of the container connecting between the first end wall and the first side wall.

13. The blank as recited in claim 12, wherein the first and second stacking tab portions each fold along a fold line, wherein the fold line is interrupted between the first and second stacking tab portions by a void separating between the first and second wall panels, so that in the container formed from the blank, the fold line is on a top edge of the stacking tabs formed of the first and second stacking tab sections, relative to the base, and wherein edges of the void form top edges of the first end wall and of the first side wall below the top edges of the stacking tabs, relative to the base.

14. The blank as recited in claim 13, wherein the first minor end flap includes a first corner panel, wherein the second minor end flap includes a second corner panel, wherein the first and second corner panels are configured to overlap and form a beveled wall connecting the first end wall to the first side wall in the container formed from the blank.

15. The blank as recited in claim 13, wherein the first minor end flap is foldably connected directly to the first wall panel, and wherein the second minor end flap is foldably connected directly to the second wall panel so that in the container formed from the blank, the first end wall is foldably connected directly to the first side wall without an intervening bevel panel.

16. The blank as recited in claim 1, wherein the first and second wall panels are configured to fold along a first fold line to overlap the first and second wall panels with one another to form the first wall panel to form a side wall, wherein the first fold line directly connects between the first and second wall panels to form a top edge of the side wall relative to the base in the container formed from the blank.

17. The blank as recited in claim 16, wherein a second fold line connects first and second portions of the stacking tab section, wherein the second fold line is spaced apart farther from the base than the first fold line, so in a container formed from the blank, the second fold line forms a top edge of a stacking tab formed from the first and second portion of the stacking tab section, wherein the second fold line is above the first fold line, relative to the base.

18. A container made from a blank as recited in claim 1, wherein each of the first side wall structures is positioned inward from the first and second wall panels.

19. A container made from a blank as recited in claim 1, wherein each of the first side wall structures is positioned outward from the first and second wall panels.

20. A container made from a blank as recited in claim 1, wherein each of the first sidewall structures is positioned inward of first minor end flaps of the second sidewall structures and outward of second minor end flaps of the second sidewall structures.
